

Li-Yue Sun

Gender : Male

Research Interests : Neural Networks, Homomorphic Encryption, Data Mining and Machine Learning

Email : john.chiang.smith@gmail.com, john.chiang.smith@hotmail.com

Status : I am available now. I used to be a third-year PhD student before I quit in May 2021.

Homepage : <https://scholar.google.com/citations?user=U78Tt8kAAAAJ>

EDUCATION & EXPERIENCE

University	Degree	Institute	Year	GPA
Xianning University	B.S.	Computer Science and Technology	2008.9 - 2012.6	~ 2.14/4
Took Chinese Postgraduate Admission Test three times			2012.6 - 2015.4	—
Tianjin University of Technology	M.S.	Computer Technology	2015.9 - 2018.3	88.53/100
Nankai University	Ph.D.	Computer Science and Technology	2018.9 - 2021.6	89.5/100
Left Nankai University after completing all the doctoral courses			2021.7 -	—

TECHNICAL SKILLS

- Programming Languages (C/C++, Python, Java, CUDA), Tools (PyTorch/TensorFlow, Eclipse, $\text{\LaTeX}$ (TikZ), C++ project by Eclipse on Linux; MyEclipse for Java EE in Windows).
- My master's thesis aims to implement a website blog system with a search engine, by the SSH frameworks (Struts, Spring and Hibernate); My bachelor's is to implement a simple BBS website system via the MVC pattern (Model-View-Controller).
- When reading the book, *Programming Collective Intelligence*, I was fascinated with Genetic Programming, also read other books in English version: *Essential C++*, *C++ Primer* and *Introduction to Data Mining*.
- I also enjoy programming some small interesting applications: I implemented a calendar-generating program and packed it into an executable file for the Windows system, which is still in use to generate a calendar. <https://github.com/petitioner/Agenda>
- I have very decent maths scores twice (136/150, 137/150) in a closed-book form test with a full math score of 150 when I took the Chinese Postgraduate Admission Test four times. I also won a Second Prize in the Advanced Mathematics Competition at the province level (Zhejiang Province) in 2010.
- IELTS: Overall 6.5 (Listening: 5.5, Reading: 8.0, Writing: 7.0, Speaking: 5.0) (April 24, 2026)
College English Test (CET-6): 518 (December 2015)
- Possess a strong understanding of the CKKS homomorphic encryption scheme and the HEAAN library, with extensive experience studying its open-source implementation.
Implemented the HEAAN scheme from scratch in Python (CPU) using Python-Flint.
Developed a GPU-accelerated implementation of HEAAN using CUDA Python (Numba).
Proficient in core homomorphic encryption techniques, including RNS, CRT, NTT, polynomial ring arithmetic, and bootstrapping.

RESEARCH WORK (first-author Preprints)

- *1st project*: **Privacy-Preserving Logistic Regression Training with A Faster Gradient Variant**
<https://arxiv.org/abs/2201.10838> <https://github.com/petitioner/IDASH2017>
- *2nd work*: **Volley Revolver: A Novel Matrix-Encoding Method for Privacy-Preserving Neural Networks (Inference)**
<https://arxiv.org/abs/2201.12577> <https://github.com/petitioner/HE.CNNinfer>
- *3rd study*: **Privacy-Preserving CNN Training with Transfer Learning: Multiclass Logistic Regression**
<https://arxiv.org/abs/2304.03807> <https://github.com/petitioner/HE.CNNtraining>